Causal Motion-Driven State Estimation and Resource-Constrained Closed-Loop Adaptation

A. SYNTHETIC EXPERIMENTAL ENVIRONMENT

Independent synthetic profile

Baseline, relaxation, stimulus gain
Motion frequency and sensitivity
Low x=2 / Elevated x=8

Simulated latent-state dynamics

Hidden continuous state x(t)
Regulated interval [4, 6]; target 5
Disturbance + stimulus-dependent evolution

Synthetic motion generation

State-dependent motion variability
3-axis position and velocity
90 Hz observation stream

Truth for audit only

Generated motion

B. CAUSAL ONLINE OBSERVATION AND DECISION PIPELINE

Motion observations

Position x/y/z
Velocity x/y/z
No hidden-state input

Completed 1-s window

90 causal samples
Latest observed sample: t-1
First action at 1.0 s

Feature extraction

14 online motion features
+ elapsed time = 15 inputs
Causal, no future samples

Frozen AI estimator

StandardScaler +
Logistic Regression
Low / Regulated / Elevated

Reserved controller

Directional steps <= 0.05
Neutral-return policy
Variation reserve <= 0.6

Only motion-derived features and time reach the estimator; simulated latent state remains privileged evaluation information.

C. ACTUATION AND CLOSED-LOOP STATE RESPONSE

Next observation

Updated motion response

Profile-specific state-motion coupling
New noisy position/velocity samples
Subsequent causal observation window

Next-state evolution

Stimulus and disturbance affect x(t)
Recovery, stability, overshoot
evaluated against simulated truth

Normalized stimulus adjustment

Single simulated stimulus u(t) in [0, 1]
0.5 neutral; stepwise adaptation
Control decisions at 10 Hz

Experiment: 60 training profiles | 20 development-validation profiles | 200 independent test profiles

Synthetic computation only: no physiological recordings, independent multimodal stimulus actuation, or physical-VR validation